One shell. Two player patterns.

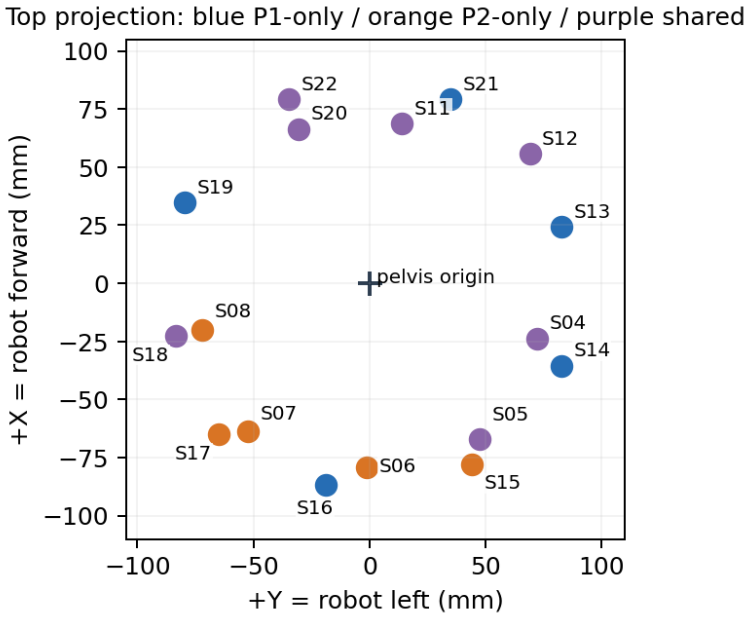
17 mounting stations / 12 reflective stickers per robot / engraved v3 shells

Use the engraved v3 halves for both players. The circle / X / circled-X marks identify installation membership; see the final page. P1 and P2 differ only in which 12 stations receive reflective stickers. Keep the other five stations nonreflective. Both assets preserve the nominal **pelvis_link** origin and ROS axes: **X forward, Y left, Z up**.

Install	Front / rear	CMTracker / Motive / body
Player 1	7 / 5	P1.json / P1.motive / UCB_P1
Player 2	4 / 8	P2.json / P2.motive / UCB_P2

P1, supplied selection: S19, S20, S11, S12, S22, S21, S13, S14, S04, S05, S16, S18.

P2, calculated selection: S04, S05, S06, S07, S08, S11, S12, S15, S17, S18, S20, S22.



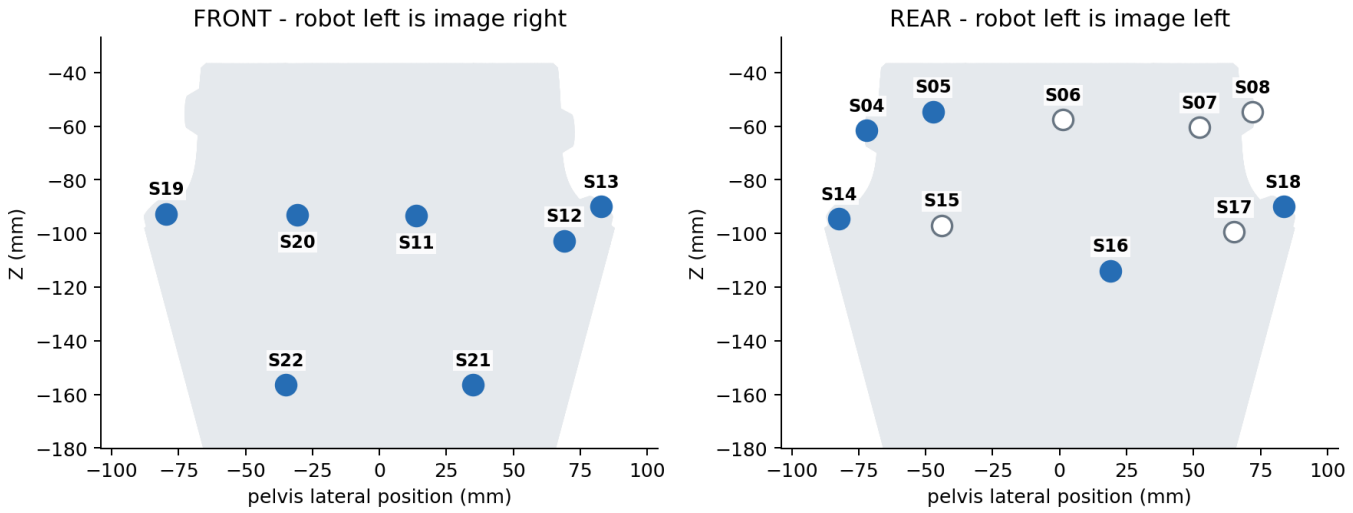
Identity limit: seven locations must be shared because 12 + 12 - 17 = 7. Shared-only observations can be identical. This is an optimized candidate under the documented CAD screening criterion, not an all-view or partial-occlusion guarantee.

P1 sticker installation

Filled circles = reflective sticker / hollow circles = nonreflective station

Install these 12: S19, S20, S11, S12, S22, S21, S13, S14, S04, S05, S16, S18.

Leave these five nonreflective: S06, S07, S08, S15, S17. Do not add reflective balls, foil, tape or bright labels at these stations.



The grey surfaces are projections of the current engraved printable STL files. Original Sxx labels identify physical stations; array position in the JSON is only authoring order. Drawings identify locations and do not prove camera visibility.

Region	Reflective stations
Front	S19, S20, S11, S12, S22, S21, S13
Rear upper	S04, S05
Rear middle	S14, S16, S18

Centre one 12 mm round sticker on each selected flat face after trial-fitting the flush pin. Film plus adhesive totals 0.20 mm. Use the same clean surface and centring method on both robots; inspect under the mocap cameras' IR illumination.

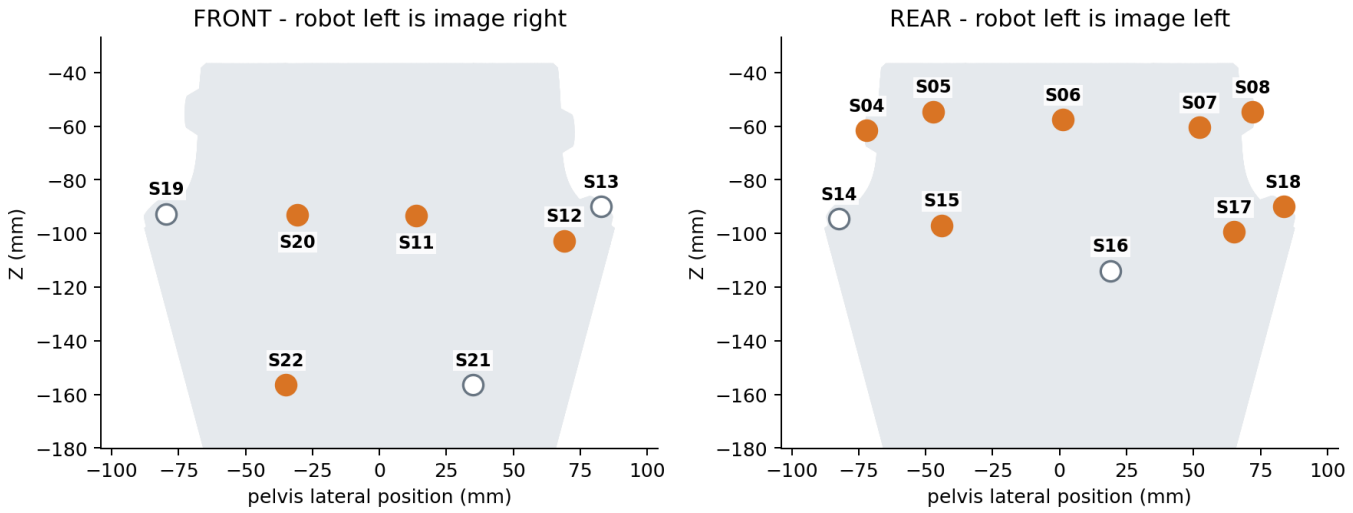
Import **vrpn_asset/P1.json** for ChingMu or **motive_asset/P1.motive** for OptiTrack. Both contain the same 12 optical centres and body name **UCB_P1**. Do not recenter the body or rotate its local frame to face the table.

P2 sticker installation

Filled circles = reflective sticker / hollow circles = nonreflective station

Install these 12: S04, S05, S06, S07, S08, S11, S12, S15, S17, S18, S20, S22.

Leave these five nonreflective: S13, S14, S16, S19, S21. Do not add reflective balls, foil, tape or bright labels at these stations.



The grey surfaces are projections of the current engraved printable STL files. Original Sxx labels identify physical stations; array position in the JSON is only authoring order. Drawings identify locations and do not prove camera visibility.

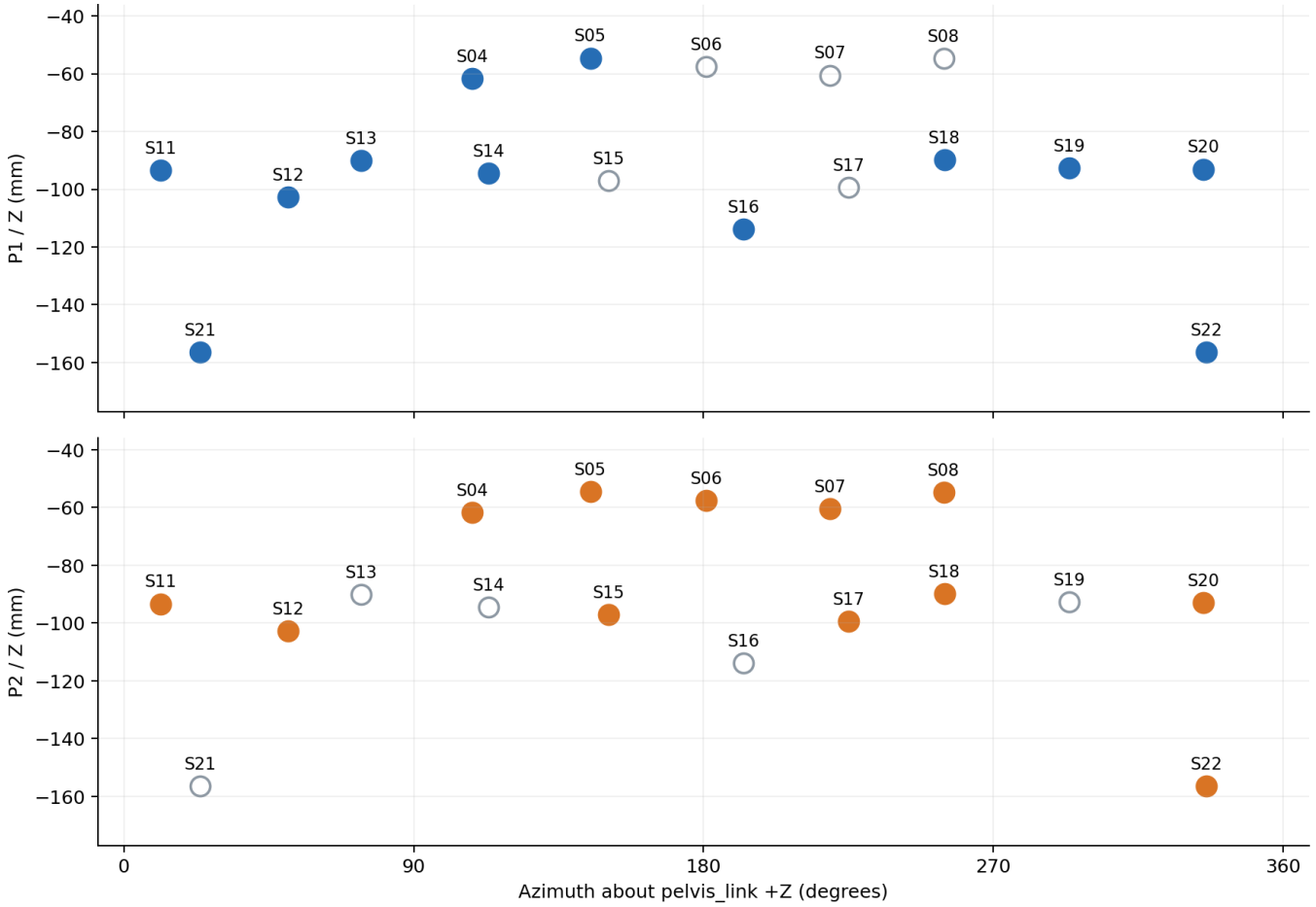
Region	Reflective stations
Front	S11, S12, S20, S22
Rear upper	S04, S05, S06, S07, S08
Rear middle	S15, S17, S18

Centre one 12 mm round sticker on each selected flat face after trial-fitting the flush pin. Film plus adhesive totals 0.20 mm. Use the same clean surface and centring method on both robots; inspect under the mocap cameras' IR illumination.

Import **vrpn_asset/P2.json** for ChingMu or **motive_asset/P2.motive** for OptiTrack. Both contain the same 12 optical centres and body name **UCB_P2**. Do not recenter the body or rotate its local frame to face the table.

Walk-around installation map

Unwrapped azimuth: front 0/360 deg / robot left 90 / rear 180 / right 270



This is an angular index, not a print template. Use it with the front/rear views and the coordinate table; do not measure sticker placement from this drawing. Both stations S21 and S22 sit lower on the front; P2 uses S22 only.

P1-only identifiers: S13, S14, S16, S19, S21.
P2-only identifiers: S06, S07, S08, S15, S17.
Shared: S04, S05, S11, S12, S18, S20, S22.

All P2-only sites are on the rear. Arrange actual camera coverage so these sites can be reconstructed despite arms and torso occlusion. A front-only observation of P2 is a subset of P1's front pattern.

Optical centres and asset order

Metres in pelvis_link / same coordinates for both players / sticker thickness included

Station	P1 index	P2 index	X (m)	Y (m)	Z (m)
S04	9	1	-0.023681	0.072053	-0.061631
S05	10	2	-0.067166	0.047266	-0.054582
S06	-	3	-0.079001	-0.001320	-0.057604
S07	-	4	-0.063806	-0.052205	-0.060498
S08	-	5	-0.019875	-0.072036	-0.054669
S11	3	6	0.068681	0.013753	-0.093372
S12	4	7	0.055718	0.068988	-0.102706
S13	7	-	0.024198	0.082666	-0.090054
S14	8	-	-0.035526	0.082456	-0.094529
S15	-	8	-0.077885	0.044045	-0.097023
S16	11	-	-0.086616	-0.018974	-0.113904
S17	-	9	-0.065030	-0.065131	-0.099352
S18	12	10	-0.022710	-0.083580	-0.089898
S19	1	-	0.034664	-0.079647	-0.092672
S20	2	11	0.066488	-0.030622	-0.093038
S21	6	-	0.079132	0.034881	-0.156415
S22	5	12	0.079113	-0.034871	-0.156403

Indices are 1-based XYZ triplet positions in ChingMu's **markerPose** array and member_id values in the authored Motive file. Native re-export may reorder labels. They are not VRPN sensor indices. A dash means leave the station nonreflective.

ChingMu stores ROS-axis coordinates in millimetres. Motive stores [X, Z, -Y] in native Y-up metres. Both preserve full CSV precision. The 17-row CSVs and workbook remain unchanged. Display rounding is not a physical accuracy claim.

The intended fixed body-to-pelvis transform is translation [0,0,0] m and quaternion [0,0,0,1] (xyzw). The streamed world-to-body pose is dynamic. Do not subtract either pattern's centroid or apply an extra P2 rotation.

Station hardware and printing

Existing flat 12 mm seat / blind screw bore / side-loaded captive nut

Feature	Existing nominal dimension
Sticker support face	12.0 mm diameter; flat, no raised rim
Support foot	20.0 mm tapered foot below optical plane
Central bore	3.4 mm diameter; 4.6 mm blind axial depth
Flush pin	3.45 mm diameter x 1.30 mm length
Sticker + adhesive	12 mm round x 0.20 mm total thickness
Nut pocket	5.8 mm across flats x 2.65 mm high
Pocket top	1.5 mm below seat; outside side access
Provisional nut	M3; 5.5 mm across flats, about 2.4 mm thick

Sticker mode: use 12 reflective stickers per robot. Trial-fit each required pin flush; nominal interference is provisional. Do not force a pin or sand the optical support plane. Unselected stations may be left bare or capped with nonreflective material; verify they do not appear as markers under IR.

Ball mode: the CAD retains its existing mounting feature, but these P1/P2 assets describe stickers, not sphere centres. Remove sticker/pin, verify the actual ball thread, nut retention and stud reach. Candidate stud penetration must not exceed 4.0 mm below the seat without a new hardware check. Do not drill through the pelvis-facing blind floor.

A 12 mm ball diameter does not establish its thread or mounted centre. Measure each ball/stud/spacer offset before authoring ball assets; do not reuse these sticker coordinates or the old 6 mm registration stand-off. Neither mode has a qualified crash-survival rating.

Print the two existing halves separately at 100% scale in millimetres. Approximate footprints are 176 x 194 mm before brim/support margins. In OpenSCAD use F5 for assembly preview; select half_a or half_b for F6/export. With bed_oriented=true each opens toward the bed at Z=0. Print placement does not change the marker coordinates.

Fit and installation checks

Engraved exterior / preserved stations, fasteners and datum

1. Inspect slicing. Check thin ribs, bridges, seams, structural screws, blind bores and side nut slots. Keep support scars off the flat marker faces. No printer/material-specific qualification is supplied.

2. Preserve the interior. The fitting interior, station geometry and structural fastener holes are preserved; only exterior marking recesses were added. Fit on a secured robot and inspect pelvis clearance, cables, ventilation, seam closure and full joint travel. Do not create new through-holes.

3. Populate the correct pattern. Use pages 2-4 and check every Sxx station twice. Each robot has exactly 12 reflective stickers. Use the engraved circle / X / circled-X legend to identify the selected stations. Keep extra markings nonreflective.

4. Validate each imported body. Confirm the 12 selected positions, pelvis origin and axes after a native export. Test the two bodies separately first, then together. Any all-17 ChingMu/Motive definition is incompatible with these 12-sticker installations.

5. Challenge occlusion. Check front, rear and side camera coverage; arms, torso, low grazing views and reacquisition after complete loss. Intentionally expose only shared stations to demonstrate the ambiguity limit. Record wrong-name matches, pose jumps and dropouts.

6. Check the physical pose. Compare the streamed pelvis pivot and local axes against physical datums and a measured displacement. Record residual error; nominal CAD coordinates are not measured calibration. Verify server world/table axes and units separately.

7. Keep the limitation visible. Six-marker tracking does not prove player identity; absence of another marker is not reliable evidence of its pattern under occlusion. Accept a result only after the actual camera setup has been checked.

Engraved mesh checks (2026-11-01): front 92,456 triangles, rear 131,182; each is one closed, consistently wound component. All 17 protected mounts, 816 top-face rays and 17 blind-bore rays pass. The smallest sampled remaining material path under engraving is 1.836 mm. Current hashes are in engraving_validation.json; MeshLab self-intersection checks were not rerun.

Historical local wall sampling found roughly 2.5 mm at most restored sites, but about 1.50 mm at an old front f1 edge. Preserve that region in slicing. Those checks are sampled evidence, not a certified global thickness bound or physical fit test. See geometry_repair_status.md and printability_validation.json.

Import P1 and P2 into ChingMu

Two files / distinct body names and IDs / identical pelvis frame convention

1. Back up the CMTracker/MCServer project. Import vrpn_asset/P1.json and vrpn_asset/P2.json into a test project. These follow the supplied CMTracker JSON schema; they are not generic VRPN import files. Do not import asset_manifest.json or player_pattern_analysis.json.

2. Confirm identities. P1.json contains UCB_P1, ID 1; P2.json contains UCB_P2, ID 2. File names are P1/P2, while the internal UCB_ prefix preserves HOPE topic routing. Resolve ID collisions before use. Asset IDs do not declare VRPN sensor indices.

3. Preserve the frame. markerPose uses millimetres in pelvis_link; Rbc remains identity. Keep X forward, Y left, Z up. Do not recenter, mirror, reset orientation to the table, apply Motive Y-up mapping or add sticker thickness again.

4. Export and compare. Confirm exactly 12 XYZ triplets and the correct station list for each body. A reorder can be equivalent, but the geometry, pivot and axes must remain equivalent. These files have offline checks, not a native import certificate.

5. Match the HOPE profile. Install the companion updated V4 profiles and select ChingMu/VRPN, V4, and the correct table side while stopped. V4 is the software alias for this v3 shell. The profile now carries a separate 12-marker definition for each side. Do not pair these imports with the old identical 17-point profiles.

6. Verify the stream. VRPN sends poses, not a ModelDef or a remotely measured asset hash. Check measured displacement to select scale 0.001 for millimetres or 1.0 for metres. Confirm the table world frame independently: Z up, tabletop Z=0, X in [0,2.74], Y in [-1.525,0] m.

7. Preserve P2 local axes. The same local coordinate convention applies to both robots. The existing runtime table-side transform handles P2's world coordinates; do not bake that 180-degree rotation into P2.json.

The inherited minMarker=6 is retained; it is not an identity guarantee. OptiTrack uses the matching P1/P2 Motive pair described next. See vrpn_asset/README.md for ChingMu commissioning and reproduction commands.

Import P1 and P2 into OptiTrack

Matching sticker subsets / native Y-up storage / ROS Z-up stream

1. Back up the Motive project. Test-import motive_asset/P1.motive and P2.motive in an isolated project. These native XML candidates have offline checks only; Motive import/re-export and physical tracking are not yet verified. Do not import the old all-17 assets for these installations.

2. Confirm two distinct bodies. P1.motive is UCB_P1, streaming ID 9; P2.motive is UCB_P2, ID 10. Each has 12 constraints, member IDs 1-12, station names Sxx and new distinct UUIDs. Resolve existing body-name and streaming-ID collisions before use.

3. Preserve the pelvis frame. All points are nominal CAD sticker centres, not a new camera measurement. Native offsets are [ROS X, ROS Z, -ROS Y] in metres. Set Streaming Up Axis to Z Up and keep modeldef_y_up_to_z_up=true in the adapter. Do not apply that conversion twice.

4. Keep identity body-to-pelvis extrinsics. Do not reset pivot location/orientation, subtract the marker centroid, add sticker thickness again, mirror P2 or bake a table-side rotation into its local coordinates. Both body-local origins represent pelvis_link with X forward, Y left, Z up after conversion.

5. Export and verify. Re-export each body; compare all 12 offsets (allow reordered labels), pivot, axes and IDs with asset_manifest.json. No learned constraint_graph was invented. If Motive rejects the candidate, use the manifest's native coordinates to author the body through supported Motive tooling, then repeat the export check.

6. Verify the ROS path. Default motive_live accepts the selected live ModelDef without a file pin. Optional motive_asset mode now checks the selected player's exact 12-point reference. This is a nominal geometry check, not proof of physical fit or identity under occlusion. The legacy measured stickers_v3 capture still requires all 17 sites and is not suitable for this pair.

7. Qualify the installation. Check measured pelvis displacement, axis directions, table/world registration and simultaneous two-body tracking. Test shared-only views, loss and reacquisition. Names and UUIDs cannot resolve identical observed subsets; these patterns have seven common stations.

See motive_asset/README.md and asset_manifest.json for hashes, coordinates and validation limits. Vendor references: docs.optitrack.com/motive/rigid-body-tracking and docs.optitrack.com/motive/data-streaming. Original measured exports remain unchanged as historical evidence, not current imports.

How P2 was selected

Exhaustive CAD screening / explicit design assumptions / partial-view limits

There are 6,188 ways to choose 12 of 17 stations. The minimum possible overlap with fixed P1 is seven, so P2 must use all five P1-absent rear sites. This leaves 792 maximum-separation candidates, all evaluated by the reproducible authoring tool.

Rank in order: (1) minimize nonshared four-point distance-signature candidates within 2 mm; (2) minimize nonshared three-point candidates at 2 mm; (3) maximize worst P2 normal-facing count; (4) maximize views containing a P1-only point; (5) maximize views with six P2-facing points, then total facing count and P1-only redundancy. A station-order tie break makes the choice deterministic.

CAD metric	Selected P2 result
Overlap / exclusive per player	7 shared / 5 unique each
Nonshared 4-point candidates, 2 mm	0 (conservative distance-signature screen)
Nonshared 3-point candidates, 2 mm	6
Same screen at 3 / 5 mm, four points	1 / 24 candidates
Same screen at 3 / 5 mm, three points	15 / 70 candidates
P2 normal-facing count, 288 views	minimum 2; mean 3.934
Views with P1-only / P2-only facing point	286 / 178 of 288

Signature screening compares sorted inter-point distances; it is a necessary rather than sufficient condition for a rigid fit. The 2 mm edge-distance threshold is a design assumption, not measured CMTracker accuracy. Larger tolerances admit more similarities. No camera layout, torso/arm occlusion, shell ray tracing or triangulation is included.

Facing grid: azimuth 0-355 degrees in 5-degree steps, elevation 0/15/30/45 degrees; outward-normal incidence at most 60 degrees. These are upper bounds on one-direction marker availability, not reconstructed marker counts. Cameras must supply multiple views. The analysis JSON also reports 45/70-degree sensitivity.

Unavoidable ambiguity: the shared seven contain 35 identical triples, 35 identical quadruples and seven identical six-point subsets. Eight genuinely observed matched stations would exclude a shared-only observation in the nominal geometry; this is not a vendor recognition guarantee, and no eight-point gate is imposed. No pattern satisfying this request can be certified unambiguous from every partial view.

CAD datum and mount-seat reference

Unchanged marker positions / original station IDs retained

The existing nominal registration is preserved: pelvis X = CAD Z / 1000, pelvis Y = CAD X / 1000, pelvis Z = (CAD Y - 145.800095037096) / 1000. The CAD source uses millimetres and CAD Y up. The marker datum is pelvis_link, not either subset's centroid.

Station	Seat X (m)	Seat Y (m)	Seat Z (m)
S04	-0.023648	0.071856	-0.061636
S05	-0.067006	0.047146	-0.054589
S06	-0.078802	-0.001319	-0.057610
S07	-0.063659	-0.052070	-0.060507
S08	-0.019844	-0.071838	-0.054673
S11	0.068482	0.013736	-0.093385
S12	0.055579	0.068885	-0.102807
S13	0.024146	0.082537	-0.090198
S14	-0.035453	0.082323	-0.094659
S15	-0.077708	0.043973	-0.097082
S16	-0.086419	-0.018963	-0.113934
S17	-0.064902	-0.065000	-0.099433
S18	-0.022665	-0.083449	-0.090042
S19	0.034583	-0.079518	-0.092801
S20	0.066293	-0.030580	-0.093060
S21	0.078934	0.034855	-0.156395
S22	0.078915	-0.034846	-0.156383

Sticker centre = seat + 0.20 mm along the outward normal. The CSVs and workbook contain complete translation, quaternion (xyzw) and normal data. Each marker-local frame maps into pelvis_link; local +Z is outward. Round-sticker roll is an authoring convention, not an optical measurement.

The registration derives from nominal CAD references, with numerical consistency checks documented in CAD_registration.md. The old measured Motive subset has 1.395 mm RMS / 1.853 mm maximum fit residual; five restored rear sites were not observed in that export. Those records neither validate the new P2 installation nor establish physical pelvis accuracy. Keep the original source exports and geometry evidence for traceability.

Engraved installation symbols

Rear wordmark / circle, X and circled X / unchanged optical geometry

The rear HITCH OPEN wordmark lies above the cooling vent and below the rear marker stations. A second line below it reads circle P1, separator dot, X P2. Small recessed symbols beside all 17 mounts identify membership without marking the flat 12 mm optical seats.

P1 only - circle

S13, S14, S16, S19, S21

P2 only - X

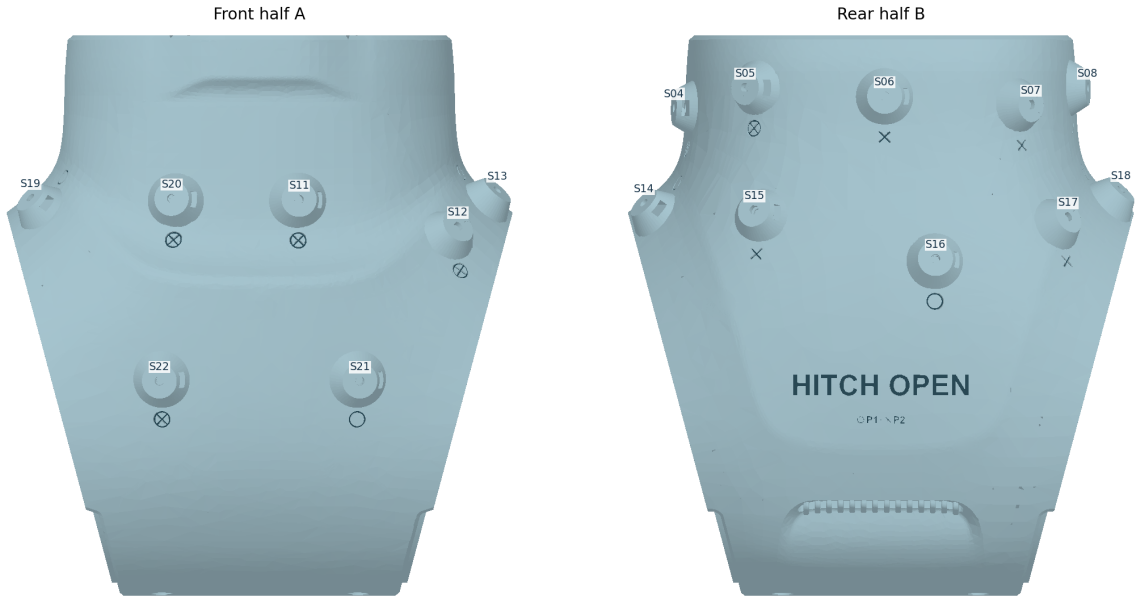
S06, S07, S08, S15, S17

P1 and P2 - circled X

S04, S05, S11, S12, S18, S20, S22

A3 pelvis shell v3 / engraved installation guide

Actual revised STL projections. Engraved floors shaded darker; station IDs are drawing annotations.



Circle: P1 only **X: P2 only** **Circled X: P1 + P2**

Station faces, blind bores, mounting geometry, cooling vent and pelvis datum are preserved.

Symbols: 6 mm across, 0.7 mm strokes, 0.4 mm recessed. Wordmark: 60.94 x 7.50 mm, 0.60 mm recessed. The second-line legend is 16.57 x 2.50 mm, 0.40 mm recessed. Most symbols lie below their mounts; S13, S14, S18 and S19 lie above to avoid thin edge regions.

These are shallow exterior cuts. Marker coordinates, sticker faces, blind bores, nut access, shell seams, cooling ventilation and the pelvis datum are preserved. The symbols do not resolve shared-only optical ambiguity. Physical fit and strength still need validation.